

AI-Powered Business Verification and Workforce Automation System

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Abstract—Business verification is a critical process for organizations that must validate the legitimacy of companies, workforce records, and supporting documentation before conducting business operations. Traditional verification methods often rely on manual review, requiring employees to compare business information across multiple sources, which can be time-consuming, inconsistent, and susceptible to human error. This project presents the AI-Powered Business Verification and Workforce Automation System, a web-based platform designed to streamline the verification process through workflow automation and centralized data management. The system combines a React-based frontend, an Express.js backend, and a PostgreSQL database to provide users with a unified dashboard for managing businesses, reviewing documents, initiating verification requests, and monitoring verification activity. When a verification request is submitted, the backend coordinates an automated verification workflow that retrieves publicly available business information, compares it with stored records, and generates verification results, confidence scores, and recommendations. These results are stored within the database and displayed through an interactive dashboard that supports efficient review and decision-making. The completed prototype demonstrates the feasibility of integrating modern web technologies with automated verification workflows to reduce manual effort, improve consistency, and provide a scalable foundation for future enhancements, including document intelligence and advanced AI-assisted verification techniques.

Index Terms—business verification, workflow automation, PostgreSQL, React, Express.js, web automation

I. INTRODUCTION

- **Problem Statement:** Organizations frequently verify business information before establishing partnerships, approving vendors, processing workforce documentation, or ensuring regulatory compliance. These verification activities commonly involve reviewing business websites, validating contact information, comparing supporting documents, and confirming that company records are accurate and current. In many organizations, these tasks remain largely manual, requiring employees to gather information from multiple systems before making verification decisions. As organizations continue to process increasing numbers of businesses and documents, manual verification workflows become increasingly inefficient. Repetitive verification tasks consume valuable employee time, introduce opportunities for inconsistent decision-making, and reduce overall operational efficiency. These challenges are particularly significant when verification

requests must be completed quickly while maintaining accuracy and accountability.

- **Motivation:** The motivation for this project was to explore how workflow automation and intelligent verification tools can improve the efficiency of business verification without eliminating human oversight. Rather than replacing decision-makers, the objective was to automate repetitive verification tasks while providing users with organized information and confidence-based recommendations that support informed decisions. The project also provided an opportunity to integrate several technologies commonly used in modern software engineering, including frontend development, backend API services, relational databases, and automated web verification. Bringing these components together into a single platform allowed the team to develop a practical solution that addresses a real-world business problem while demonstrating full-stack system integration.
- **System Contribution:** The AI-Powered Business Verification and Workforce Automation System provides a centralized platform for managing businesses, organizing verification records, and automating portions of the verification workflow. Users can create and manage business records, review supporting documentation, initiate verification requests, and monitor verification activity through a unified dashboard. The system integrates a React frontend with an Express.js backend and a PostgreSQL database using REST APIs. Verification requests trigger an automated workflow that retrieves business information, compares it with stored records, generates confidence scores, and stores the resulting verification history within the database. By combining business management, verification automation, and workflow tracking into a single application, the project demonstrates how modern web technologies can improve consistency, reduce manual effort, and provide a scalable foundation for future intelligent verification systems.

II. RELATED WORK

Business verification is something many organizations perform before working with a company, hiring vendors, or processing important records. In many cases, employees still have to manually visit company websites, search public databases, and compare information from different sources to determine whether a business is legitimate. While this process

can be effective, it is often slow, repetitive, and can lead to mistakes when large numbers of businesses need to be reviewed.

Some existing business verification tools automate parts of this process, but many focus on identity verification or financial compliance rather than providing a complete workflow. They often verify information but do not help users organize businesses, track verification requests, or manage the overall process.

Our project takes a different approach by combining business management, workflow automation, and automated verification into one system. Instead of making users perform every verification step manually, the application gathers publicly available business information, compares it with stored records, calculates a confidence score, and displays the results in one dashboard. This helps reduce repetitive work while still allowing users to make the final decision.

III. METHODOLOGY

A. Overall System Architecture

The AI-Powered Business Verification and Workforce Automation System was designed using a modular full-stack architecture that separates the user interface, application logic, data management, and automated verification services into independent components. This modular design improves maintainability, simplifies future development, and allows each component to perform a dedicated function while communicating through well-defined REST APIs.

The overall workflow begins when a user interacts with the React-based web application to submit a new business, upload information, or request a business verification. These requests are sent to the backend server through HTTP API calls, where they are processed before interacting with the database and verification services. This layered architecture separates presentation, business logic, and data storage, creating a scalable foundation for future enhancements.

The backend was built using Node.js and Express.js to handle the main functions of the system. It connects the frontend, database, and verification process. When a user starts a verification, the backend receives the request and uses Playwright with Chromium to visit the company's website and collect business information. The system then compares the information it finds with the business record and generates a confidence score based on how closely they match.

Finally, the backend sends the results to the frontend, where users can view the verification status, confidence score, and other details on the dashboard. The backend is organized into separate modules, making it easier to maintain and improve in the future.

The database uses PostgreSQL as its relational database management system to provide a centralized and structured repository for all business verification data. The database was designed using a normalized schema to reduce redundancy and maintain data integrity through primary and foreign key relationships. It stores business information, verification records, workflow tasks, verification queue entries, notifications, and activity logs, allowing the application to track the entire

verification lifecycle. The database is integrated with the rest of the system through APIs.

B. Frontend

- The frontend of the AI-Powered Business Verification and Workforce Automation System was developed using React to provide users with a responsive and interactive interface for managing the verification process. React was selected because of its component-based architecture, allowing the application to be organized into reusable interface elements while supporting efficient state management and client-side navigation.
- The application consists of several primary views, including the dashboard, businesses, document center, verification, and reports pages. React Router enables navigation between these pages without requiring full page reloads, providing a smoother user experience. Each page focuses on a specific aspect of the verification workflow while sharing common components such as navigation controls, status cards, data tables, and verification summaries.
- The businesses page allows users to create, edit, search, and delete business records stored within the PostgreSQL database. User input is validated before being transmitted to the backend through REST API requests, ensuring that business information remains consistent throughout the system.
- The verification page provides an interface for initiating automated business verification. When users select a business for verification, the frontend submits a request to the backend API, which executes the verification workflow and returns the resulting verification data. Once the response is received, the interface displays website verification status, contact verification results, confidence scores, and recommended actions while simultaneously updating the verification history displayed throughout the application.
- The dashboard aggregates information retrieved from multiple backend endpoints and presents key performance indicators, including total businesses, pending verifications, completed verifications, and recent verification activity. This centralized view allows users to monitor system activity and quickly identify outstanding verification requests.
- Communication between the frontend and backend occurs through asynchronous REST API calls using the Fetch API. This architecture separates presentation logic from business logic, allowing the user interface to remain responsive while backend services perform verification tasks and database operations independently.

C. Backend

The backend of the AI-Powered Business Verification and Workforce Automation System was built using Node.js and Express.js. It manages the business verification process and connects the frontend with the database.

- **Express Server:** Handles requests from the frontend and manages communication between the different parts of the system.
- **API Routes:** Receive requests from the frontend, process the verification, and return the results.
- **Crawlee:** Automates the process of collecting publicly available business information from company websites.
- **Chromium:** Works with Crawlee to open and browse company websites during the verification process.
- **Company Verification Workflow:** The system visits a company's website, collects business information, and compares it with the information stored in the database.
- **Confidence Scoring:** Generates a confidence score based on how closely the collected information matches the business record.
- **Verification Results:** Once the verification is complete, the backend sends the results back to the frontend, where they are displayed on the dashboard for users to review.

D. Database

- PostgreSQL was selected as the project's relational database management system because it provides reliable data storage, strong relational capabilities, and seamless integration with the Node.js/Express backend. It serves as the centralized repository for all business verification information.
- The database was designed using a normalized relational schema consisting of tables for businesses, verification records, workflow tasks, verification queue entries, notifications, and activity logs. This structure reduces data redundancy while maintaining consistency across the application.
- Primary and foreign key relationships connect businesses with their verification history, workflow tasks, notifications, and activity logs. These relationships preserve data integrity and allow the system to efficiently retrieve related information during the verification process.
- The businesses table serves as the central entity within the database. As businesses progress through the verification workflow, the system automatically updates business statuses, stores verification results, creates workflow tasks, generates notifications, and records activity logs.
- The PostgreSQL database communicates with the backend through SQL queries executed using the `pg` library. REST API endpoints perform create, read, update, and delete operations while also retrieving the data displayed on the frontend dashboard.
- The database supplies real-time information to the dashboard, including business records, verification history, workflow progress, notifications, and system statistics. This centralized design enables workflow automation while providing a scalable foundation for future enhancements and production deployment.

IV. RESULTS

The system was tested to make sure the verification process worked correctly. During testing, the backend successfully retrieved business information from company websites. The frontend displayed the verification results, including the business information, verification status, and confidence score. The verification results were also stored in the PostgreSQL database, allowing users to view previous verification records. Overall, the testing showed that the frontend, backend, and database worked together successfully.

V. DISCUSSION

The AI-Powered Business Verification and Workforce Automation System demonstrates that automating the business verification process can reduce manual effort while improving organization and workflow management. The combination of a React frontend, Express backend, and PostgreSQL database created a platform for storing business information, managing verification requests, tracking workflow progress, and presenting real-time statistics through an interactive dashboard.

One of the strengths of the project is its modular architecture, which separates the frontend, backend, database, and verification components. This design improves maintainability and allows additional functionality to be integrated with minimal impact on existing features. Another strength is the relational PostgreSQL database, which provides a structured and scalable approach for storing businesses, verification history, workflow tasks, and system activity.

Even though this project has strengths, there are several limitations. The application is a prototype that has primarily been evaluated using sample business records and controlled testing scenarios. Some business verification requests still require manual review when sufficient information is unavailable or verification confidence is low. The verification process also depends on the availability and accuracy of publicly accessible business information, which may be inaccurate or unavailable.

The riskiest assumption in this project is that publicly available business information provides sufficient evidence to accurately verify a business through an automated process. The evaluation partially tested this assumption by using synthetic business records while executing the full verification pipeline against real websites and storing the resulting verification data, workflow updates, and dashboard statistics. This demonstrated that the system's workflow and automation function correctly. However, the evaluation did not validate the accuracy or reliability of the verification process across a large and diverse set of real businesses. Additional testing with real-world datasets and broader business scenarios would be required before the system could be considered ready for production use.

VI. CONCLUSION

This project successfully developed an AI-Powered Business Verification and Workforce Automation System that automates business verification while improving workflow management through a centralized web application. The system integrates

a PostgreSQL database, backend APIs, workflow automation, verification tracking, notifications, and a real-time dashboard to reduce repetitive administrative tasks and improve the organization of business verification activities. The completed prototype demonstrates that combining artificial intelligence with workflow automation can significantly improve the efficiency, consistency, and transparency of business verification processes.

The project has considerable potential and can be further improved in the future. Possible enhancements include incorporating additional business data sources, expanding workflow automation with more intelligent task routing and background processing, enhancing dashboard analytics and reporting capabilities, implementing role-based authentication and user management, and deploying the system as a scalable cloud-based application capable of supporting larger organizations and production-level workloads. The project demonstrates that the concept works and that its effectiveness could be increased through these enhancements.